Measuring Spatiotemporal Sampling Sensitivity in Video Action Recognition: A Cross-Architecture Analysis at Scale

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Abstract. Temporal resolution in video action recognition is usually treated as a fixed hyperparameter rather than a sampling constraint. We present a large-scale study of empirical classifier-evidence loss under temporal and spatial subsampling across 8 architectures and 8 datasets. We first characterise what a strided sampling protocol actually varies: because strided candidates are re-uniformised to a model's fixed frame budget, stride alters the input only once the candidate pool falls below that budget. The resulting sensitivity therefore measures how much distinct evidence a model receives, not the sampling rate of the underlying signal, and it is gated by the ratio of clip length to frame budget (Spearman 0.96 against dataset-level sensitivity). Controlling for this by supplying every architecture the same distinct frames at the same positions, models still differ in accuracy per distinct frame; at $k = 2$ distinct frames, the matched accuracy ranking remains associated with the original ordering (Spearman 0.86, $p = 0.007$), though agreement is weaker at other budgets and the previously reported 3–5× magnitude does not survive. We introduce the **Temporal Demand Score (TDS)** to quantify accuracy loss under temporal subsampling; its ranking is exactly preserved by the curve-integral and unclipped variants, and largely preserved after baseline normalisation. In an input-scale/resampling stress test, CNN native checkpoints are more sensitive than the evaluated Transformer and SSM checkpoints; we do not interpret this as zero-shot spatial-grid robustness. We separate the gain from target-resolution adaptation (≈32pp) from the gain due to additional training alone (≈7pp). Furthermore, hardware profiling shows video state-space models are 4–8× slower than efficient Transformers. A maximum-probability confidence cascade traces a lower-frame operating point, which we report with held-out calibration as a deployment illustration rather than a mitigation method. Finally, we provide a reproducibility package and an interactive visualization dashboard, and we release the diagnostic that identifies, for any dataset and frame budget, the stride at which a fixed-budget pipeline degenerates.

Keywords: Temporal sampling · Video recognition · Evidence loss

1 Introduction

Applications of activity recognition include driving, healthcare, and surveillance [2, 13]. The field has evolved from 3D ConvNets and dual-pathway net-

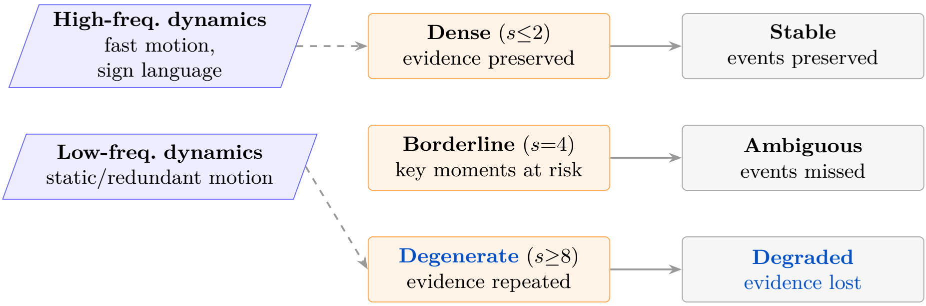

Fig. 1: Temporal evidence loss: sparse sampling can miss or distort the frames a recognizer needs, especially for temporally localized actions.

works [4, 8, 29, 31, 35] to video Transformers and state-space models [1, 3, 7, 15, 18, 27, 33].

Despite this diversity, all models are evaluated using standardized sampling recipes (8–32 uniformly-sampled frames at fixed stride), treating temporal resolution as a static implementation detail. This assumption fails in real-world deployments where constraints on computational hardware, power usage, sensor capabilities, or communication bandwidth make dense sampling infeasible [17]. This leaves two fundamental questions unanswered: *when* does temporal subsampling become destructive, and *why* do different architectures respond differently?

Sampling perspective. The Nyquist–Shannon theorem [23] motivates the intuition that undersampling can distort a temporal signal rather than merely compress it, a principle also relevant to perception [32] and deep learning [38]. We initially adopted this as a conceptual guide. We now state a sharper limitation, because it shapes how every temporal result below must be read: when a recogniser consumes a fixed number of frames, the realised sampling rate is that number divided by the observed span, so no frame-subsampling protocol can vary sampling rate without simultaneously varying the evidence available or the temporal context. Aliasing in the signal-processing sense is therefore *not identifiable* from such a sweep, and we do not claim to measure it. What we measure, and what the TDS metric quantifies, is empirical classifier-evidence loss under controlled subsampling. We test this interpretation through sampler diagnostics, a fixed-frame-count span sweep, matched-evidence architecture comparisons, and multi-stride training (Sec. 4.2; supplementary material).

Contributions. We conduct a systematic cross-architecture study of spatiotemporal sampling sensitivity:

1. **Temporal Demand Score (TDS)**: an architecture-averaged metric that measures a dataset's sensitivity to reduced temporal evidence. We demonstrate its utility by ranking 8 diverse datasets with leave-one-architecture-out analysis, architecture-family ablation (CNN-only, Transformer-only), and bootstrap confidence intervals checking ranking stability (Sec. 4.1; supplementary material).

2. **Cross-architecture sampling-sensitivity characterization**: The 8,000-configuration sweep reveals a frame-budget confound; after all architectures receive the same distinct frames at the same positions, their evidence-efficiency ordering remains correlated with the original ordering (Spearman 0.857, $p = 0.007$), but the claimed 3–5× margin does not (Sec. 4.2).
3. **Target-resolution fine-tuning**: CNNs show the largest off-native penalty. A same-resolution continued-training control attributes approximately 32pp of the reported 39pp recovery to target-resolution adaptation and approximately 7pp to additional training alone (Sec. 4.5).
4. **Measured inference latency**: GPU latency characterization across all 8 models × 5 target resolutions (bfloat16), enabling sampling-aware deployment budgeting per device class (Sec. 4.6).
5. **Practical illustration and reproducibility package**: A maximum-probability cascade is evaluated with held-out threshold calibration. It matches dense TimeSformer accuracy on SSv2 at 6.9 average frames (-0.03pp, 95% CI $[-1.07, +0.83]$), but does not generalise cost-free across all model–dataset pairs; we therefore present it only as a deployment illustration. We also provide a supplementary package containing a deterministic dashboard, evaluation scripts, and result tables; public code will be released after review (Sec. 4.7).

2 Related Work

2.1 Video Recognition Architectures

CNN-based architectures. 3D ConvNets learn spatiotemporal features end-to-end [4, 29], while Non-local Networks introduce global self-attention [35]. Factorized variants such as R(2+1)D and MC3 improve optimization and efficiency [30, 31]; SlowFast processes two frame-rate streams simultaneously [8].

Transformer and SSM architectures. Video Transformers extend patch-based recognition to time: TimeSformer uses divided space-time attention [3], ViViT uses factorized attention [1], and VideoMAE learns via masked patch reconstruction [27, 33]. VideoMamba [15] brings selective state-space scanning [6, 11] to video. Prior robustness work benchmarks spatial and photometric corruptions [21]; we instead isolate temporal and spatial sampling effects across this architectural spectrum.

2.2 Temporal Sampling and Adaptive Inference

Efficient inference methods sample or process fewer frames. TSN and TSM define sparse-sampling recipes [17, 34]; AdaFrame selects frames [37]; AdaFocus crops regions [36]; AR-Net adapts resolution [20]; and FrameExit uses confidence cascades [9]. Dataset-level temporal reasoning has been studied with shuffled frames [12] and motion-only cues [22]. These works do not measure the accuracy cliff induced by controlled stride and coverage sweeps; our TDS metric provides an architecture-averaged temporal-demand ranking.

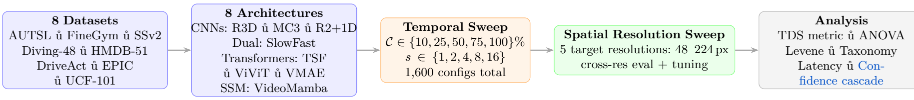

Fig. 2: Pipeline: the core factorial sweep uses 8 architectures × 8 datasets × 5 target resolutions (48–224px) × 5 coverages × 5 strides = **8,000 configurations**.

Table 1: TDS = mean accuracy drop (stride=1→16, cov=100%) per Eq. 1, averaged over all 8 architectures (n=8). Sorted by TDS. **Bold** = the two co-equal highest-demand domains (their relative order is leave-one-out sensitive; ranks 3–8 are stable under the supplementary robustness analysis).

Dataset	Classes	Videos	Domain	TDS	n
FineGym [24]	97	1,960	Fine-grained sport	**55.9pp**	8
AUTSL [25]	226	4,418	Sign language	**53.0pp**	8
SSv2 [10]	174	3,464	Causal/temporal	27.6pp	8
DriveAct [19]	33	448	In-vehicle	21.9pp	8
Diving-48 [16]	48	743	Fine-grained	19.2pp	8
HMDB-51 [14]	51	885	Sports	16.6pp	8
EPIC-Kitchens [5]	89	1,020	Egocentric	9.7pp	8
UCF-101 [26]	101	2,020	Appearance	4.9pp	8

2.3 Sampling Theory in Deep Learning

Sampling theory predicts that undersampling a signal below its highest temporal frequency creates aliasing [23]. Temporal aliasing is well documented in perception [32]; in deep learning, strided operators can violate shift-invariance unless anti-aliased [38]. Frequency-domain analysis has been explored for action recognition [28], but not as a large-scale cross-architecture stress test of modern video models.

3 Methodology

3.1 Datasets and Architectures

Table 1 summarizes the 8 datasets selected to span semantic domains, motion frequencies, and recording conditions. We evaluate 8 architectures from four families: (i) **3D CNNs**: R3D-18, MC3-18 [30], R(2+1)D-18 [31]; (ii) **dual-pathway**: SlowFast-R50 [8]; (iii) **Transformers**: TimeSformer [3], ViViT [1], VideoMAE [27]; (iv) **SSM**: VideoMamba [15]. All models are fine-tuned from Kinetics-400 checkpoints using AdamW with cosine decay. We use fixed validation clip lists across models (approximately 20 clips/class, or all available clips for smaller classes); additional split, crop, seed, and frame-normalization details are in the supplementary material.

3.2 Temporal Sampling Protocol

We decouple *temporal coverage* $\mathcal{C}$ (fraction of clip observed) from *temporal stride* s (sampling density), forming a 5×5 grid: $\mathcal{C} \in \{10, 25, 50, 75, 100\}\%$ and $s \in$

$\{1, 2, 4, 8, 16\}$. The temporal sweep at native resolution yields $8 \times 8 \times 25 = \mathbf{1{,}600}$ model–dataset–configuration triples. Repeating this grid at the 5 target resolutions from 48–224px yields the **8,000** core configurations.

Coverage implementation. Coverage is applied from the *beginning* of each clip: the first $\lfloor T \cdot \mathcal{C}/100 \rfloor$ frames form the candidate pool, and s decimates it. This simulates a sensor that observes only a leading fraction of the action and gives the sharpest measurement of how early the model has enough temporal context. Random window placement would confound temporal structure with missing-data effects.

Each architecture consumes a fixed frame budget B: after decimation, candidate frames are re-uniformised to B, or padded by repeating the final frame if too few remain. Thus the sweep measures distinct evidence reaching the model, not sampling rate alone; the full sampler, interval-in-seconds, and shortfall diagnostics are in the supplementary material.

3.3 Spatial Resolution Protocol

Cross-resolution evaluation. The resolution grid is 48, 96, 112, 160, and 224 px. For this no-fine-tuning branch, we first square-resize decoded frames to the requested resolution and then evaluate the unchanged native checkpoint with its stored preprocessing. Thus this is an *input-scale/resampling* stress test; it does not instantiate Transformers on a new positional-token grid or interpolate positional embeddings. Native training resolution is 112px for CNNs and 224px for Transformers and VideoMamba.

Target-resolution fine-tuning. For each model–dataset–resolution triple at the 5 target resolutions (48–224 px), we fine-tune from the native-resolution checkpoint for 10 additional epochs. For Transformers and VideoMamba, positional embeddings are bicubically interpolated from the native grid to the target grid before fine-tuning. CNNs transfer directly, and SlowFast uses adaptive average pooling to support non-native resolutions.

3.4 Temporal Demand Score (TDS)

We define TDS as the architecture-averaged accuracy drop from stride=1 to stride=16:

$$\mathrm{TDS}(\mathcal{D}) = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \big[\mathrm{Acc}(m, \mathcal{D}, s{=}1) - \mathrm{Acc}(m, \mathcal{D}, s{=}16)\big]^{+} \tag{1}$$

where $[x]^{+} = \max(x, 0)$, so a small negative drop (sparse sampling slightly outperforming dense sampling) contributes 0 rather than reducing the dataset score. We define feature collapse operationally as stride=1 Top-1 accuracy below 5%; no model–dataset pair in the native-resolution TDS computation met this criterion, so no entries are excluded and the ranking is unchanged. TDS is architecture-pool averaged, but its dataset ranking is stable under leave-one-architecture-out, family ablations, and bootstrap resampling over our 8-model pool (Sec. 4.1; supplementary material), suggesting it reflects dataset temporal structure rather than any single model.

3.5 Statistical Analysis

Because every cell evaluates the same clips, we use clip-level repeated-measures ANOVA with coverage and stride as crossed within-subject factors and report partial η^2 for coverage, stride, and their interaction. Only clips present in all 25 cells enter each balanced analysis. Levene's test detects per-class variance inflation under sparse sampling. Post-hoc Welch tests with Bonferroni correction identify the stride at which accuracy falls sharply.

3.6 Maximum-Probability Cascade

Given temporal-sweep results at two operating points—cheap: cov=100%/s=4 (4 effective frames); dense: cov=100%/s=1 (16 frames)—we route video v by threshold:

$$\hat{y}(v) = \begin{cases} \hat{y}_{\text{cheap}} & \text{if } \max_c p(c \mid v, s{=}4) > \tau \\ \hat{y}_{\text{dense}} & \text{otherwise} \end{cases}$$

The rule thresholds the maximum class probability, not the predictive entropy; we previously called it entropy routing, which was a misnomer, and refer to it here as a *maximum-probability cascade*. We sweep $\tau \in [0.10, 0.95]$ on a calibration half of the validation split and score the selected τ on the held-out half, repeating over 1000 stratified splits so the reported gain carries an interval. No retraining is required; only the existing temporal-sweep inference is reused.

4 Results

4.1 Temporal Demand Score and Ranking Robustness

Figure 3 (left) ranks datasets by TDS. **FineGym (55.9pp)** rivals AUTSL sign language (53.0pp), showing that sport is not inherently low-TDS when labels depend on brief phase-specific evidence. Under the evaluated protocol, both datasets lose substantially more distinct classifier evidence than the remaining domains. UCF-101 (4.9pp) is nearly appearance-driven: object identity and static scene cues often suffice.

Crucially, ranks 3–8 are stable under leave-one-architecture-out. One leave-one-out pool swaps only the #1/#2 positions, so we report FineGym and AUTSL as co-equal high-TDS domains. All model–dataset pairs use the same protocol; the swap reflects estimator sensitivity. The stability also holds for CNN-only, Transformer-only, and Transformer+SSM pools (Spearman $\rho \geq 0.976$ against the full pool), while bootstrap resampling gives a FineGym−AUTSL 95% CI of [−4.6, +11.7]pp, consistent with zero (supplementary material).

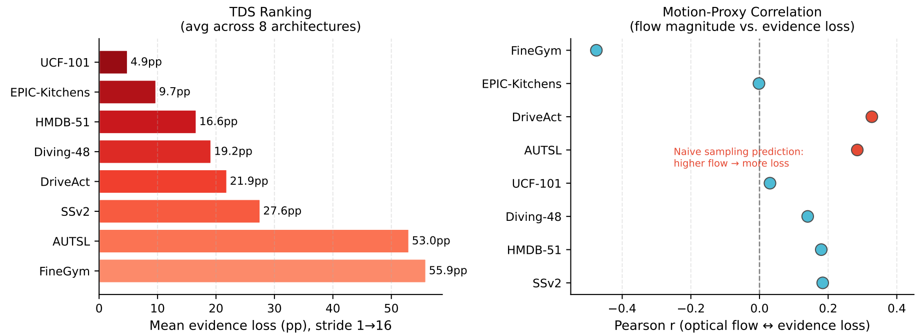

Fig. 3: Left: TDS ranking—mean accuracy drop from stride=1 to 16 averaged over all 8 architectures (n=8). FineGym (55.9pp) and AUTSL (53.0pp) are co-equal highest-demand domains (order between them is leave-one-out sensitive; see supplementary material); UCF-101 (4.9pp) is appearance-dominated. Positions 3–8 are exactly stable under leave-one-architecture-out. **Right**: Pearson correlation between optical-flow magnitude and per-class evidence loss (8 datasets). AUTSL and DriveAct show a positive correlation; EPIC-Kitchens is near-zero (background camera motion confounds flow); FineGym shows the strongest correlation in the opposite direction ($r = -0.475$): discrete-event classes lose more evidence than sustained-motion classes under this protocol.

Exploratory motion-proxy analysis. Figure 3 (right) correlates optical-flow magnitude with per-class evidence loss. AUTSL and DriveAct show a positive within-dataset trend: classes with more motion lose more accuracy. EPIC-Kitchens is near zero, suggesting that egocentric camera motion decouples raw flow from action-relevant demand. FineGym is inverted ($r = -0.475$, $p < 0.0001$): high-flow gymnastics classes often remain visually distinctive from almost any frame, while low-flow held positions or balance corrections depend on brief key moments. Thus raw flow magnitude is only a partial proxy for temporal demand, especially when class identity hinges on a discrete event rather than sustained motion (Table 2).

The inversion generalizes beyond FineGym. A spectral check on all 8 datasets across 5 resolutions pairs whole-frame optical-flow cutoff frequency with empirical stride-sensitivity at the same resolution (40 points; supplementary material). The relationship is significantly *negative* ($\rho = -0.63$, $p = 1.1{\times}10^{-5}$): UCF-101 and HMDB-51 have high flow-derived cutoffs but low stride-sensitivity, while FineGym and AUTSL have the highest stride-sensitivity at lower cutoff frequencies. Thus whole-frame flow does not validate a literal Nyquist account; it shows that classifier-relevant evidence is often localized and diluted by bulk motion. A supplementary sliding-window diagnostic is more consistent with this evidence-localization view: per-class correct-class confidence concentration is positively associated with per-class evidence loss (pooled Spearman $\rho = +0.26$ over a 6-architecture ensemble).

Table 2: Exploratory motion-proxy analysis: Pearson r between whole-frame optical-flow magnitude and per-class evidence loss. Mixed signs show that flow magnitude is not a literal proxy for temporal demand.

Dataset	Classes	Pearson r	Significant	Interpretation
FineGym	97	-0.475	✓	Discrete-event classes lose more evidence
DriveAct	33	0.327	marginal	Flow ∼ arm motion
AUTSL	226	0.285	✓	Flow ∼ hand speed
SSv2	174	0.184	✓	Causal/object motion
HMDB-51	51	0.180	×	Mixed motion types
Diving-48	47	0.140	×	Body rotation dominated
UCF-101	101	0.031	×	Appearance dominates
EPIC-Kitchens	89	-0.002	×	Camera motion confounds

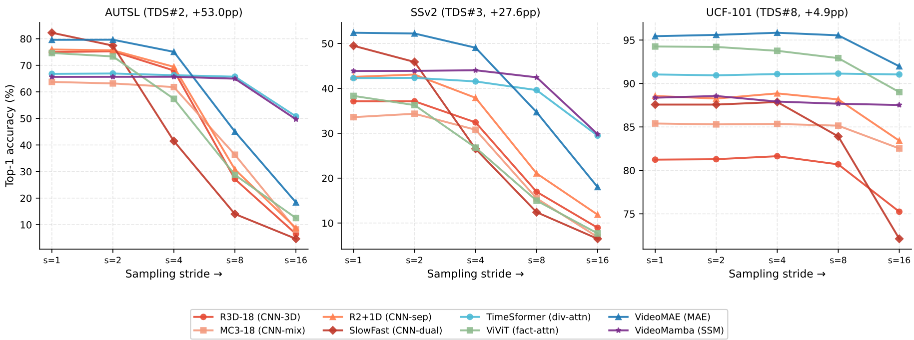

Fig. 4: Top-1 accuracy under the original fixed-budget stride protocol (cov=100%) for representative high-, medium-, and low-TDS datasets. Curves are descriptive: their drop combines distinct-frame loss with architecture and frame-budget effects; matched-evidence results are reported in Sec. 4.2.

4.2 Cross-Architecture Temporal Sampling Sensitivity

The raw comparison is confounded by unequal native frame budgets. With every architecture given the same k distinct frame positions ($k \in \{1, 2, 4, 8\}$), matched accuracy agrees with the original ranking only at $k = 2$ (Spearman 0.857, $p = 0.007$); other budgets are weaker. We therefore retain the curves as descriptive and do not attribute their original 3–5× spread to temporal aggregation. Full matched-evidence results are in the supplementary material.

Table 3 is retained as a descriptive account of the original protocol, not as a causal architecture comparison. Its raw drops combine model design, native frame budget, clip length, and the sampler's repeat-last behaviour. The matched-evidence experiment above is the basis for the narrower architecture claim.

Cross-architecture sensitivity heatmap. Figure 5 shows the stride-induced accuracy drop (stride=1 minus stride=16, in percentage points) at cov=100%, with

Table 3: Full temporal sensitivity table: stride=1 accuracy (%) / drop at stride=16 (pp), cov=100%. Rows sorted by average drop across all 8 datasets.

Model (family)	UCF-101	SSv2	HMDB-51	Diving-48	FineGym	AUTSL	DriveAct	EPIC-Kit.	Avg drop
TimeSformer (div. attn)	91.0/0.0	42.3/12.8	79.9/4.0	38.2/2.0	66.4/36.1	66.8/16.0	67.6/8.5	32.3/2.9	**10.3pp**
VideoMamba (SSM)	88.4/+0.8	43.9/14.1	69.8/3.6	36.5/4.9	72.7/40.8	65.6/16.0	57.6/9.6	28.3/1.2	**11.4pp**
MC3-18 (CNN-mix)	85.4/2.9	33.6/26.7	78.6/9.7	31.6/12.4	64.2/51.1	63.7/55.5	69.0/13.8	36.2/8.6	22.6pp
R3D-18 (CNN-3D)	81.2/6.0	37.1/28.2	80.3/21.1	28.8/16.0	71.5/61.4	75.0/68.5	68.3/23.9	37.2/12.7	29.7pp
R2+1D (CNN-sep.)	88.6/5.1	42.6/30.7	73.1/17.6	35.3/18.8	76.8/64.2	75.9/67.2	62.5/24.8	35.5/11.7	30.0pp
ViViT (fact. attn)	94.3/5.2	38.3/30.6	80.2/19.4	53.0/36.3	69.9/55.2	74.6/62.1	67.4/20.3	32.9/11.1	30.0pp
VideoMAE (MAE)	95.4/3.5	52.4/34.4	84.0/20.2	48.6/23.4	78.0/67.8	79.6/61.2	73.9/40.8	37.7/10.8	32.8pp
SlowFast (dual)	87.6/15.4	49.5/43.0	79.3/37.4	50.5/39.7	78.2/71.5	82.3/77.6	72.5/33.7	39.4/18.5	**42.1pp**

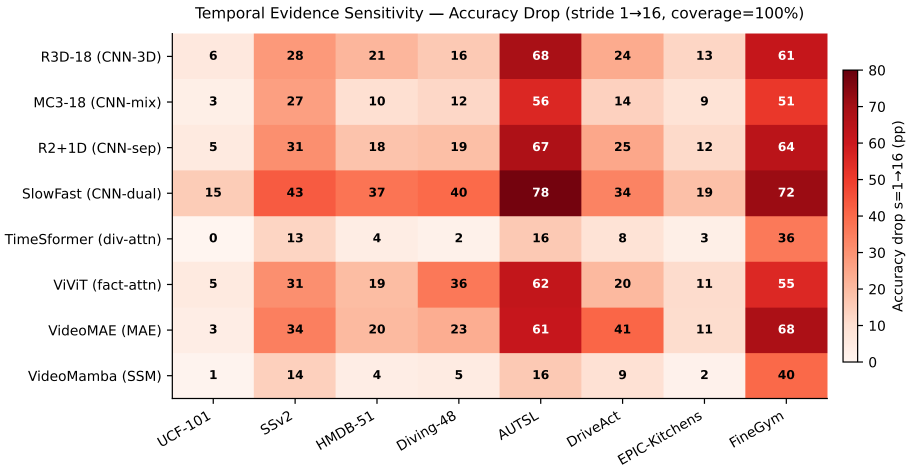

Fig. 5: Cross-architecture evidence-loss heatmap: accuracy drop (percentage points) from stride=1 to stride=16 at cov=100% under the original fixed-budget protocol. Models are sorted by increasing mean drop. The cells show drop, not Top-1 accuracy, and should be interpreted together with the matched-evidence control.

models sorted by mean drop. This is an *accuracy-drop* heatmap, not a Top-1 accuracy heatmap; larger values indicate stronger evidence loss. Full per-model, per-dataset Top-1 accuracy heatmaps across coverage and stride are in the supplementary material.

4.3 Statistical Validation (ANOVA and Levene's Test)

Effect sizes. Table 4 reports the corrected within-subject analysis. Coverage remains the largest mean effect, but by a smaller margin than the independent-cell analysis implied: partial η^2 is 0.292 ± 0.152 for coverage and 0.232 ± 0.145 for stride, a ratio of 1.26× rather than 2.2×. Their interaction is non-negligible at 0.070 ± 0.081 and reaches 0.271 on SlowFast/AUTSL. Among models with complete clip-level grids, the descriptive stride-effect ordering is unchanged (TimeSformer 0.124 to SlowFast 0.331). Full per-dataset results are provided in the supplementary material.

Table 4: Clip-level repeated-measures analysis of the coverage×stride grid. Values are partial η^2 (mean±std over 56 model–dataset analyses with complete per-clip grids).

Within-subject effect	Partial η^2	Interpretation
Coverage	0.292 ± 0.152	Largest mean effect
Stride	0.232 ± 0.145	Substantial effect
Coverage×stride	0.070 ± 0.081	Non-negligible interaction

Table 5: Taxonomy tier boundaries (accuracy drop, pp): classes below the Low threshold lose little accuracy; classes above the High threshold are strongly affected. Computed at cov=100%, stride=1→16.

Dataset	Low ≤	High >	# High	# Low
AUTSL	66.9pp	73.9pp	75	74
Diving-48	25.0pp	47.7pp	16	16
SSv2	59.7pp	76.9pp	58	57
HMDB-51	8.4pp	28.5pp	17	17
EPIC-Kitchens	0.0pp	19.4pp	30	11
DriveAct	14.8pp	51.0pp	11	11
UCF-101	1.4pp	7.5pp	35	31

Variance inflation. Levene's test shows significant inter-class variance inflation in 67% of model-dataset pairs ($p < 0.05$, Bonferroni-corrected across all 64 model×dataset comparisons). The worst cases are VideoMAE/HMDB-51 (2.0×) and R3D-18/HMDB-51 (1.85×), while TimeSformer and VideoMamba remain stable ($\approx 1.1\times$). Thus sparse sampling can create class-specific failures that mean accuracy hides.

4.4 Action Sensitivity Taxonomy

We classify each action class into three tiers by per-class evidence loss (accuracy drop, stride=1→16). Tier boundaries use the 33rd/67th percentiles.

High-sensitivity actions lose the most evidence under the evaluated protocol: in AUTSL, even the Low tier loses 38pp. In contrast, the UCF-101 Low tier loses only 0.3pp: static, appearance-driven actions are virtually unaffected. Table 5 reports per-dataset tier boundaries. AUTSL's Low threshold starts at 66.9pp, above the High threshold of any other dataset, placing sign language in a distinct empirical evidence-demand regime.

Clip duration effect. Counter-intuitively, shorter clips lose more accuracy (Pearson $r = -0.3$ to -0.8 across models): short clips have less temporal redundancy, so each missed frame removes a larger fraction of available evidence. On UCF-101, clips >6s lose only 1.5pp at stride=16 vs. 16.2pp for clips <3s. This suggests that short clips should receive dense sampling irrespective of model confidence (details in the supplementary material).

4.5 Spatial Resolution and Target-Resolution Fine-Tuning

Input-scale sensitivity of native checkpoints. Figure 6 and Table 6 show the effect of pre-resizing source frames before evaluation by the unchanged native

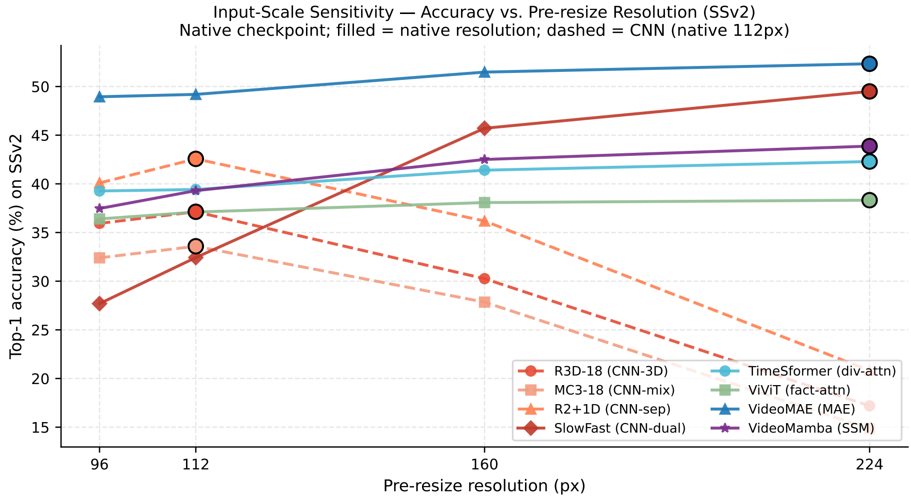

Fig. 6: Input-scale sensitivity on SSv2: source frames are pre-resized to the stated resolution and evaluated by the unchanged native checkpoint. This is an input-scale/resampling stress test, not zero-shot transfer to a new Transformer grid. CNNs (dashed, native=112px) degrade at off-native resolutions; Transformers and VideoMamba (solid, native=224px) stay within ±7pp across 96–224px. Filled markers denote each model's native training resolution.

checkpoints. CNNs (native 112px) are more sensitive to this input-scale shift: R2+1D falls from 42.6% at 112px to 20.8% at 224px (−21.8pp). Transformers and VideoMamba (native 224px) remain within ±7pp across 96–224px. These results characterize robustness to this resampling protocol, not zero-shot transfer across new Transformer token grids.

The same input-scale pattern appears across the 8 datasets (supplementary material): the largest CNN drop in the 48–224px pre-resize grid is 62.8pp on DriveAct, whereas Transformers remain within ±8pp on 7 of 8 datasets; EPIC-Kitchens is the main exception. This comparison does not identify the source of the difference between architectures.

Target-resolution fine-tuning recovers CNN performance. Table 7 shows family-level averages after target-resolution tuning. CNNs recover sharply: at 224px, input-scale accuracy is 30.0%, but fine-tuning reaches 69.2% (+39.2pp). The result shows that target-resolution adaptation can absorb a large input-scale shift; it does not establish that Transformer grids are intrinsically resolution-invariant.

4.6 Inference Latency and Deployment Budgeting

Table 8 reports measured bfloat16 latency at batch 1. VideoMamba is slowest (8–15ms), 4–8× slower than VideoMAE (1.6–2.8ms) despite its SSM design. VideoMAE gives the best Transformer accuracy per ms, while CNN latency scales predictably with resolution.

Table 6: Input-scale sensitivity on SSv2 under source-frame pre-resizing and unchanged native checkpoint preprocessing (no fine-tuning). Bold = native resolution. $\Delta_{\max}$ = largest drop from native.

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	35.9	**37.1**	30.3	17.2	−19.9pp
MC3-18	112px	32.4	**33.6**	27.8	14.8	−18.8pp
R2+1D	112px	40.1	**42.6**	36.2	20.8	−21.8pp
SlowFast	224px	27.7	32.4	45.7	**49.5**	−21.8pp
TimeSformer	224px	39.3	39.4	41.4	**42.3**	−3.0pp
ViViT	224px	36.4	37.1	38.1	**38.3**	−1.9pp
VideoMAE	224px	48.9	49.2	51.5	**52.3**	−3.4pp
VideoMamba	224px	37.5	39.3	42.5	**43.9**	−6.4pp

Table 7: Target-resolution fine-tuning: family-level mean accuracy (%) averaged over all 8 datasets, after fine-tuning from the native-resolution checkpoint for 10 epochs at each target spatial resolution. Each family is shown with (FT) and without (—) target-resolution fine-tuning, as requested, so the benefit is readable directly. **Note** that fine-tuning at a model's *native* resolution, where there is no resolution shift to adapt to, already gains +6 to +10.6pp for 7 of 8 models: the off-native benefit should be read as ≈32pp of resolution adaptation plus ≈7pp attributable to the additional epochs alone.

Family	Tuning	48px	96px	112px	160px	224px
CNN (3 models)	—	24.6	58.5	60.2	51.8	30.0
	FT	49.3	64.9	67.7	68.5	69.2
Transformer (3)	—	50.3	60.5	61.3	63.0	63.6
	FT	36.6	56.3	60.3	66.9	71.4
SSM (VideoMamba)	—	17.7	39.7	41.4	46.2	49.1
	FT	31.5	41.2	45.1	54.5	59.7
Dual-CNN (SlowFast)	—	5.9	35.1	42.9	62.3	67.5
	FT	36.2	73.3	71.9	71.6	67.5

4.7 Confidence-Cascade Illustration

Table 9 compares the maximum-probability cascade at average frame budget ≤ 8 (full curves in the supplementary material). We treat the cascade as a deployment illustration rather than a separately trained method, and report it under held-out calibration. AdaFocus [36] and AR-Net [20] use ResNet-50 and are informational. [9] FrameExit is the closest same-backbone reference, but remains contextual because it was reported under a different training and evaluation protocol.

With held-out calibration, SSv2/TimeSformer matches dense inference at 6.9 average frames (−0.03pp, 95% CI $[-1.07, +0.83]$). Across 56 model–dataset pairs, however, the mean change is −3.5pp; we therefore present the cascade only as a calibrated deployment illustration.

Across the full held-out analysis, the cascade is model–dataset dependent: its mean change from dense inference is −3.5pp over 56 pairs. We therefore do not claim a general accuracy-neutral cost reduction.

Table 8: Inference latency (ms/clip, bfloat16, batch=1, RTX PRO 6000 Blackwell). VideoMamba is 4–8× slower than VideoMAE despite being an SSM. Measured on a single GPU; we no longer extrapolate to other device classes, as those factors were estimates rather than measurements.

Model	48px	96px	112px	160px	224px
R3D-18	0.61	0.89	1.20	1.76	2.78
MC3-18	0.61	0.85	1.18	1.84	3.15
R2+1D	1.39	2.21	2.54	4.10	5.77
SlowFast	3.09	3.11	3.11	3.13	4.10
TimeSformer	3.25	3.24	3.20	3.20	4.30
ViViT	1.55	1.77	1.99	3.19	6.35
VideoMAE	1.57	1.79	1.78	1.92	2.82
VideoMamba	8.06	8.15	8.18	10.03	15.44

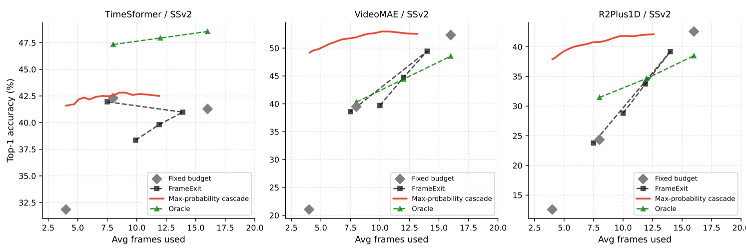

Fig. 7: Maximum-probability cascade on SSv2: descriptive in-sample accuracy–frame curve as τ sweeps between 4-frame and 16-frame inference. The held-out estimate is reported in Table 9.

5 Discussion

Architecture and sampling. Unequal native frame budgets confound the original groups. Under matched evidence, only the $k = 2$ ranking agrees with the original ordering; this remains an observational association, not an attention-type mechanism.

Spatial robustness. CNN checkpoints are more sensitive to source-frame pre-resizing than the evaluated Transformer and SSM checkpoints. This input-scale test does not identify the architectural cause; target-resolution adaptation can reduce the penalty (Table 7).

Deployment implications. Resolution, latency, and temporal evidence should be searched jointly rather than treated as independent knobs. A raw stride value is not transferable across datasets or models: the relevant boundary is where the candidate pool falls below the model's frame budget and repeated frames replace distinct evidence. This boundary can be computed from clip length, coverage, and frame budget before deployment. Target-resolution tuning makes CNN resolution a cost dial; the confidence cascade can reduce frames on favourable pairs, but requires held-out calibration and has no cost-free guarantee. These trade-offs are deployment-relevant because a model that is slightly less accurate at dense

Table 9: SSv2 comparison with the cascade threshold selected on a stratified calibration half and scored on the held-out half (1000 resamples). The same-backbone comparison to FrameExit is contextual because its training and evaluation protocol is externally reported; AdaFocus/AR-Net use a different backbone.

Method	Avg frames	SSv2 Top-1	Backbone
Informational only: different backbone			
AdaFocus [36]	8.0	54.7%	ResNet-50
AR-Net [20]	8.0	48.6%	ResNet-50
Contextual reference: same TimeSformer backbone			
Oracle upper bound	7.7	47.3%	TimeSformer
Fixed 16f (dense)	16.0	42.3%	TimeSformer
Fixed 8f	8.0	42.3%	TimeSformer
Cascade (held-out)	**6.9**	**42.2%**	TimeSformer
Fixed 4f (cheap)	4.0	41.5%	TimeSformer
FrameExit [9]	9.9	38.4%	TimeSformer

sampling can become preferable once resolution, evidence budget, and measured hardware latency are optimized together.

Limitations. Our architecture analysis is observational: temporal aggregation design covaries with tokenization, pretraining objective, capacity, input length, native resolution, and optimization recipe, so we report empirical associations rather than a controlled causal mechanism. A stronger test would swap temporal modules inside a shared backbone while holding training and capacity fixed. As Sec. 1 states, literal signal aliasing is not identifiable from our fixed-budget frame sweep. Consistent with that limitation, whole-frame flow is inversely related to stride-sensitivity, while temporal concentration of correct-class confidence is positively associated with per-class evidence loss (pooled Spearman $\rho = +0.26$; supplementary material). Our cascade estimate uses held-out calibration, but the broad model–dataset analysis still shows that confidence routing is not uniformly accuracy-neutral.

6 Conclusion

We presented a large-scale cross-architecture study of spatiotemporal sampling sensitivity in video action recognition. Across 8 architectures, 8 datasets, and 8,000 configurations, the original fixed-budget curves reflect both model behaviour and the distinct-frame evidence supplied by the sampler; matched evidence weakens the previously claimed 3–5$\times$ architecture gap. The TDS metric identifies FineGym and AUTSL as co-equal high-demand domains, establishing fine-grained sport as a high-evidence-demand regime. VideoMamba shows mixed sensitivity in the descriptive protocol and is slow on GPU; target-resolution fine-tuning closes the CNN input-scale gap by up to 39pp; and a held-out maximum-probability cascade matches dense TimeSformer on SSv2 at 6.9 average frames, but is not uniformly accuracy-neutral.

Practically, the study makes the evidence budget explicit: the onset of repeated-frame padding can be computed from clip length, coverage, and frame budget before deployment. This helps distinguish failures caused by missing observations from those caused by the recogniser when allocating frames, resolution, and latency.

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